

System Architecture

JSPL EAF Industrial AI Decision Support — Release 1.0
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System Architecture — JSPL EAF Industrial AI Decision Support

Overview

Unified platform exposing frozen research artifacts (Phases 19–32) through a single Next.js frontend and FastAPI backend.

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Operator UI (Next.js v3)

- /predict → Phase 19 model (frozen)
- /optimize → Phase 20.2 optimizer (frozen)
- /hybrid/evaluate → Phase 32 trust framework (read-only bridge)
- /optimize/v2 → Phase 31 V2 (research, planning-safe)
- /validation → Plant result recording
- /feedback → Operator review (research store)

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Layers

Layer	Technology	Phase
Prediction	CatBoost ensemble + Phase 21 feature pipeline	19
Production optimizer	Physics-guided search	20.2
Research optimizer	Multi-objective, no POWER changes	31
Trust / consensus	Hybrid decision engine	32
Explainability	Post-hoc SHAP + narratives	29
Product integration	Website + validation APIs	33

Data flow

- Operator enters planning recipe (HM, DRI, flux, shift).
- Backend normalizes inputs and runs frozen `predict_recipe()` — never retrained in Phase 33.
- Optional `hybrid/evaluate` loads Phase 32 engine via `research_bridge_service` (sys.path bridge only).
- Optimizer page can run Phase 20.2 and Phase 31 V2 in parallel for comparison.

Deployment

- Backend: `backend/` FastAPI on port 8001
- Frontend: `steelops-ai/frontend_v2/` Next.js
- Research code: `research/phase_*` — imported read-only, not copied into pickles

Non-goals (Phase 33)

- No model retraining
- No pickle or weight changes
- No modification to Phase 19/20/31/32 core modules